



CALPHAD MODELING

MatCalc steel database
9 Fe-base alloy systems

phase fractions



STRATIFIED SPLIT

cluster-stratified 64 / 16 / 20
KMeans $k = 6$, 3 seeds

train / val / test



SURROGATE TRAINING

simplex-constrained MLP + gated head
ridge, k-NN, XGBoost, forest

$$\sum f_i = 1$$

fitted surrogates



EVALUATION

interpolation, band holdout,
extrapolation; ideal-form control

phase fractions at (x, T)

DEPLOYMENT

experimental anchor and
two high-throughput screens

held-out rows